Basanite Product Overview V2.0

#Basanite
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1. Executive Summary

Basanite is an intelligent recruitment and interview system that transforms how organisations evaluate technical talent. By leveraging advanced AI capabilities, our platform generates customised interview frameworks that adapt to each candidate's background while maintaining rigorous, consistent evaluation standards grounded in validated occupational science.

This version of the product overview incorporates two foundational design updates following external academic review and psychometric consultation: the formalisation of our interview methodology as **Construct-Templated Adaptive Interviewing (CTAI)**, and the restructuring of our evaluation dimensions into a two-layer architecture anchored in ONET occupational taxonomy. Together, these changes address the four goals that guide every design decision at Basanite: interview standardisation, evaluation methodology rigour, legal and scientific defensibility, and the ability to assess genuine capability rather than interview preparedness.

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2. Defining the Problem Space

2.1 The Measurement Gap

The current talent evaluation landscape for high-complexity roles in the tech industry is largely failing. Up until April 2026, these methods typically include algorithmic coding tests, system design interviews, structured behavioural interviews, work samples and take-home projects, and conversational interviews. They are based on paradigms and frameworks invented a century ago, where human ability to work was neither augmented nor weakened by the ability to leverage AI. On the surface, these methods appear objective and scalable. In practice, they select for the misleading proxy of interview preparedness, and are structurally blind to the qualities that actually drive performance in a world where AI handles an expanding share of routine technical execution.

With the AI revolution transforming talent needs at a fundamental level, this measurement gap is producing both organisational and social consequences that compound as AI capabilities advance.

For corporations, hiring processes optimised for the wrong signals produce systematic mismatches between role requirements and the people placed into them. These are invisible at the point of hire but manifest downstream in slower execution, higher management overhead, and accelerating attrition. When evaluation frameworks cannot reliably identify who will drive product velocity, navigate ambiguity, or lead effective human-AI collaboration, organisations lose their ability to build the capability portfolios that competitive advantage depends on.

For society, as AI increasingly handles execution-layer tasks, the skills that generate human economic value are shifting toward judgment, synthesis, ethical reasoning, and collaborative intelligence. Yet talent pipelines continue to screen for the skills AI is replacing fastest, while filtering out the human qualities that remain irreplaceable. Over time, this produces a workforce allocation problem of significant scale, where individuals with genuinely differentiated capabilities are sorted into the wrong roles or excluded from high-value work entirely.

We are therefore in urgent need of a critical reassessment of talent evaluation theory and an innovative reinvention of talent evaluation paradigms and tools.

There is a further problem specific to the current moment that legacy tools have no mechanism to address: the systematic degradation of fixed-question interview validity by AI-assisted candidate preparation. Interview question databases on Glassdoor, Reddit, and practitioner communities mean that any sufficiently used structured interview instrument becomes, over time, a test of preparation rather than capability. Basanite is designed from first principles to be robust to this failure mode.

2.2 Human Intelligence That Matters

Research spanning cognitive science, philosophy of knowledge, behavioural decision theory, and organisational psychology converges on a different account of what drives genuine value in technical organisations. The qualities that distinguish high performers in complex, adaptive, AI-era work environments are not the ones that conventional hiring instruments capture. They include:

- **Judgment under ambiguity**: the capacity to act decisively on incomplete information, without either paralysis or false confidence
- **The ability to surface and transmit tacit knowledge**: what Michael Polanyi described as the irreducible gap between what experts know and what they can say, the practical intelligence that lives in experience rather than in text

- **Sound intuition in the absence of sufficient data**: what Gary Klein's decades of naturalistic decision-making research revealed as the recognition-primed judgment that distinguishes genuine experts from those who merely know the vocabulary of expertise
- **Psychological safety and collective learning**: the social and epistemic conditions under which teams can correct their own errors before they compound
- **Creative problem reframing**: the ability to recognise when the team is solving the wrong problem
- **Ethical reasoning**: not the ability to articulate ethical frameworks, but the capacity to feel the weight of real tradeoffs and navigate them with integrity
- **The capacity to be genuinely transformed by experience**: the metacognitive self-awareness that separates those who accumulate experience from those who learn from it

These qualities share a structural feature that explains why conventional hiring instruments cannot detect them: they cannot be retrieved from a knowledge base. They are not facts to be recalled or algorithms to be executed. They are capabilities forged through real experience, expressed through real judgment, and legible only to evaluators who know what to look for and how to ask.

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3. Design Philosophy

Assess genuine capability, not performed competence.

The central conviction behind every design decision at Basanite is simple: most hiring tools are optimised for the wrong signal. They measure how well someone can approximate the idea of a good candidate, rather than whether they actually are one.

3.1 Five Principles

3.1.1 Depth Over Breadth

Basanite is designed around progressive excavation: each layer of the assessment exists to move one level deeper into signal quality. A candidate who answers fluently at the surface should encounter ground that shifts beneath them at the next layer. The edges of real ability are blurry. Performed ability has no edges.

3.1.2 Standardisation as a Fairness Mechanism

Unstructured evaluation produces advantage for those who are most culturally legible to the evaluator. Standardised evaluation built on fixed question sets encourages coaching and rewards those who perform. Basanite takes a third path: standardisation at the construct level, with personalisation at the surface level.

Every Basanite question is generated from a pre-specified construct template: the cognitive demand, follow-up logic, and scoring rubric are identical for every candidate. What varies is the contextual content drawn from the candidate's own CV and stated experience. This is the principle of parallel test forms applied to interviews: just as a cognitive ability test may exist in Form A and Form B, consisting of different items, identical construct, identical norming, considered the same instrument for validity purposes. Basanite generates what is in effect a distinct but psychometrically equivalent form for each candidate.

This approach is more equitable than fixed-question structured interviews in one important respect: it prevents the advantage that accrues to candidates who have rehearsed the specific question set. Candidates who mirror the interviewer's background, communication style, or preparation access are advantaged in conventional structured interviews not because they are more capable, but because they prepared for the right variant of the question. Basanite's construct templates make this form of gaming structurally impossible.

By anchoring every evaluation to ONET-grounded dimensions, consistent scoring rubrics, and explicit BARS criteria grounded in observable evidence, the system creates conditions where a self-taught engineer without institutional pedigree can be seen as clearly as one from a target university.

3.1.3 The Encounter as Part of the Assessment

The experience a candidate has inside a Basanite evaluation should feel like a rigorous but genuinely curious conversation rather than an interrogation, a performance, or a test of how well they have prepared for exams. This matters for two reasons:

1. The collaborative, uncertainty-tolerant environments that the best technical organisations are building require people who can think under genuine pressure, not people who can perform competence in artificial conditions. An evaluation system that rewards the latter is selecting against the former.
2. Candidates who feel genuinely seen regardless of outcome become the most powerful distribution channel available. The reputation of an evaluation process travels through the talent communities it touches. A process that candidates respect is one that attracts the candidates worth reaching.

3.1.4 Honest About What AI Can and Cannot Do

Basanite will never claim more than the evidence supports. The system is strongest where AI has a genuine structural advantage over human evaluation: in consistency, in scale, in the ability to follow a thread across a long conversation without fatigue or drift, and in the capacity to maintain identical standards regardless of the order in which candidates are seen. The tenth candidate in a Basanite evaluation receives the same quality of assessment as the first. This is not true of human-led processes, and the difference matters at scale.

There are things that require irreducible human judgment, such as nuances of seniority assessment, domain-specific technical verification, and the final hiring decision itself. Basanite is designed to be explicit about these limits: flagging where human expertise is required, producing quotable evidence rather than opaque scores, and positioning itself as the infrastructure that makes human judgment better rather than the mechanism that replaces it.

3.1.5 Evaluation as a Two-Way Mirror

The best hiring processes leave candidates with a clearer understanding of themselves rather than just a verdict. Every evaluation strategy deployed by Basanite must be one that can be honestly explained to the candidate it is applied to. Strategies that depend on deception to function that work only because the candidate does not know what is being measured, are excluded not merely on ethical grounds but on epistemic ones. An evaluation that cannot withstand transparency is not producing the signal it claims to produce.

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4. Assessment Methodology: Construct-Templated Adaptive Interviewing (CTAI)

This section documents Basanite's core interview methodology in sufficient technical detail for product teams to understand, implement, and defend it.

4.1 The Methodological Problem CTAI Solves

Industrial-organisational psychology identifies two separable dimensions of interview structure, established by Campion, Palmer and Campion (1997):

- **Content structure**: standardisation of what is asked, whether every candidate receives identical questions
- **Evaluation structure**: standardisation of how responses are scored, whether identical rubrics, criteria, and follow-up logic are applied across candidates

Both dimensions independently improve predictive validity. Critically, the research indicates that evaluation structure is the more validity-predictive of the two. This has a direct implication for Basanite's design: maximum evaluation structure, achieved through pre-specified BARS rubrics, quotation-anchored scoring, and consistent follow-up logic, delivers the primary validity benefit even when content structure is not achieved at the item level.

Traditional structured interviews (Level 4 in the Huffcutt-Arthur classification) achieve standardisation at the surface by asking identical questions. This approach has two failure modes in the current environment. First, identical question sets are systematically leaked: Glassdoor, Reddit, and AI-assisted preparation tools mean that any widely deployed fixed question set becomes a preparation test rather than a capability test. Second, identical generic questions cannot be anchored to the candidate's own experience, which reduces ecological validity. A question like "tell me about a time you dealt with ambiguity" tests the candidate's ability to retrieve and narrate an appropriate memory, not their ability to actually exercise judgment under ambiguity.

CTAI resolves both failure modes.

4.2 The CTAI Principle: Parallel Test Forms

The conceptual foundation of CTAI is the parallel test forms principle from psychometrics. In cognitive ability testing, Form A and Form B of the same instrument contain different items but are considered equivalent instruments because they elicit identical constructs under identical scoring. A candidate who takes Form A and a candidate who takes Form B can be compared because the measurement is construct-level, not item-level.

CTAI applies this principle to interviews. Rather than generating a single fixed item to which all candidates respond, Basanite generates, for each evaluation dimension, a construct-equivalent question personalised to the candidate's own stated experience. "Tell me about the architectural decision you made on [CV_PROJECT] when the requirements were still uncertain" and "Tell me about the point in [CV_PROJECT_2] where you had to commit to a technical direction without complete information" are psychometrically parallel: they elicit identical cognitive constructs under identical BARS rubrics. Where traditional structured interviews have Form A, CTAI generates effectively unlimited parallel forms, one per candidate, without sacrificing the construct-level standardisation that makes comparison valid.

This is a methodologically distinct approach that preserves the validity-critical property, namely consistent construct elicitation and evaluation, while eliminating the validity-degrading property of fixed question sets: their susceptibility to preparation and gaming.

4.3 Implementation: Anatomy of a CTAI Question

Every Basanite interview question is generated from a **construct template** composed of three immutable components and one variable component:

Immutable components (identical across all candidates for this dimension):

1. **Construct scenario**: the cognitive demand the question is designed to elicit (e.g., judgment under ambiguity, tacit knowledge surfacing, decision under resource constraint)
2. **BARS rubric**: the 1–5 scoring scale with behaviourally anchored descriptors at each level, derived from ONET occupational descriptors for the relevant dimension
3. **Follow-up logic**: the mandatory depth sequence and the trigger conditions that govern whether verification probes are deployed

Variable component (personalised from candidate CV): 4. **Context slot**: the specific project, role, or experience named in the question, drawn from the candidate's uploaded CV

The question generation process fills the context slot from the CV. It does not generate the construct scenario, the rubric, or the follow-up logic, these are pre-specified by the dimension architecture and are invariant.

Example:

Construct template: "Describe a moment in [CONTEXT] where you had to make a consequential technical decision without sufficient data to feel confident about it. What did you do, and why did you not wait for more information?"

Candidate A receives: "Describe a moment during your work on the data pipeline migration at [Company X] where you had to make a consequential technical decision without sufficient data to feel confident about it..."

Candidate B receives: "Describe a moment during the API redesign you led at [Company Y] where you had to make a consequential technical decision without sufficient data to feel confident about it..."

The questions are different at the surface. The construct elicited, the rubric applied, and the follow-up logic are identical. Both candidates are measured with the same instrument.

4.4 ONET-Based Dimension Auto-Recommendation

When a hirer inputs a job description, Basanite's configuration agent automatically maps the role to a recommended combination of ONET evaluation dimensions. This mirrors the logic of Criteria Corp's role-battery recommendation system, but grounded in the richer ONET occupational taxonomy rather than Bureau of Labor Statistics job titles alone.

The recommendation engine analyses the job description for signals including: role title and seniority indicators, technical domain and stack, scope of ownership (individual contributor vs. technical lead vs. architect), and contextual signals about the organisation (stage, team size, ambiguity tolerance). It matches these against Basanite's dimension taxonomy to produce a recommended evaluation set with a plain-language rationale for each dimension.

Hirers can:

- Accept the recommended set as a complete package
- Remove dimensions they consider less relevant to this specific role
- Browse the full dimension library and add dimensions not included in the default recommendation

The interface presents each dimension with a plain-language description, the ONET code it maps to, and example signal indicators, so the hirer does not need to understand the underlying assessment methodology to make informed choices.

4.5 Follow-Up Question Logic

This part needs to be refined on further research!

Follow-up questions in Basanite serve a categorically different purpose from follow-up questions in a conventional interview. They are not information-gathering tools. They are **construct verification probes**: deployed specifically when the initial answer raises doubt about whether the capability being claimed is genuine.

This distinction has a practical implication for interview design: follow-up probes are not applied uniformly. Most candidates who give genuine, specific, friction-containing answers to the core template questions will complete the dimension without triggering verification probes. Candidates whose answers raise authenticity flags encounter additional depth. This produces a more efficient interview for strong candidates while ensuring that potential false positives receive proportionally more scrutiny.

Trigger conditions for follow-up deployment:

A follow-up probe fires when any of the following signals are detected in the candidate's response to a core template question:

- **Comprehensive coverage without prioritisation**
- **Absence of concrete parameters in technical claims**
- **Clean narrative arc without friction**
- **Textbook framing**
- **Low signal confidence after full response**

Verification probe selection:
The specific probe deployed is matched to the trigger type rather than drawn from a generic follow-up bank:

Trigger	Probe technique
Comprehensive coverage without prioritisation	Forced Commitment: demand a specific, defensible position
Absence of concrete parameters	Parameter Verification: ask for exact numbers, sizes, timelines
Clean narrative without friction	Regret Probing or Honest Failure Elicitation
Textbook framing	Fourth Wall Break: ask what actually came to mind first, before they organised their answer
Low signal confidence	Progressive Excavation: go one level deeper into the same territory
Implausibly confident technical claim	Boundary Condition Probing: ask under what conditions the approach fails

5. Evaluation Dimension Architecture

Basanite's evaluation dimensions are structured into two layers, each held to an appropriate and explicit evidential standard.

5.1 Layer 1: ONET-Anchored Core Dimensions

These dimensions form the basis of the candidate's scored evaluation and composite ranking. Each is mapped to a validated ONET work activity code, with BARS rubrics derived from ONET occupational descriptor language. Construct validity is established by inheritance from validated occupational science infrastructure rather than requiring bespoke psychometric validation from scratch.

Dimension	ONET Anchor	Description
Judgment under ambiguity	4.A.2.b.4 Judgment and Decision Making	The capacity to act decisively on incomplete information without paralysis or false confidence
Sound intuition under data scarcity	4.A.2.b.2 Making Decisions and Solving Problems; Gary Klein Recognition-Primed Decision model	Recognition-primed judgment that distinguishes genuine expertise from vocabulary-level familiarity
Creative problem reframing	4.A.3.b.1 Thinking Creatively	The ability to recognise when the team is solving the wrong problem
Role-specific technical depth	Role-family-specific ONET skill descriptors	Calibrated to one of two depth registers (applied or research/architecture) based on job description; measured through boundary condition probing and parameter verification

Additional Layer 1 dimensions may be recommended by the configuration agent for specific role families (e.g. Coaching and Developing Others for technical lead roles, Information Ordering for data-intensive roles).

Scoring standard: No dimension may receive a score above 3 without a specific verbatim statement from the candidate cited as evidence. Scores are grounded in what was actually said, not in overall impression.

5.2 Layer 2: Observation Layer

These observations are generated in parallel with the scored evaluation and included in the hirer report as contextual information. They do not contribute to the candidate's composite score or ranking position. They are provided to support the quality of the human final interview.

Current Layer 2 observations include the capacity to be genuinely transformed by experience, a construct with strong theoretical grounding in metacognitive research but insufficient psychometric validation history to include in scored evaluation at this stage.

Report language for Layer 2: All Layer 2 content is clearly labelled in the hirer report: *"The following observations are provided for context only. They do not form part of this candidate's evaluation score and should not be used as criteria in your hiring decision. They are provided to support the quality of your final conversation."*

Legal and disclosure requirements for Layer 2: Candidates are informed during onboarding that the hirer may receive additional contextual observations not reflected in their evaluation score. Hirer agreements include explicit contractual language prohibiting the use of Layer 2 observations as hiring criteria. The platform maintains an audit log confirming that Layer 2 output is structurally separated from the ranking algorithm. Adverse impact monitoring (see Section 10.3) includes analysis of whether Layer 2 observations correlate with demographic variables in practice.

5.3 Cheating Risk as an Independent Dimension

Cheating risk is scored and reported separately from capability across both layers. A candidate who may have used AI assistance but produced strong responses is treated differently from one who produced weak responses. The two assessments do not collapse into each other. This distinction is critical: Basanite is not claiming that AI assistance automatically disqualifies a candidate, rather, it provides the hirer with the information needed to interpret the assessment output accurately.

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6. Prototype Architecture

The high-level system architecture comprises five components:

Configuration agent: processes the hirer's job description, performs ONET role-family mapping, generates the recommended dimension set and construct templates, and activates the role-specific interview agent prompt.

Interview agent: conducts the candidate assessment using CTAI methodology. Maintains signal confidence monitors for each active dimension. Deploys core template questions from CV-filled construct templates. Triggers verification probes when authenticity flags are raised. Tracks narrative consistency across the full conversation.

Scoring engine: applies BARS rubrics to candidate responses in real time. Requires verbatim citation for scores above 3. Generates dimension scores, cheating risk assessment, and technical capability map. Produces composite ranking score.

Report generator: produces hirer report and candidate report from the same assessment data, with Layer 2 observations included in the hirer report under explicit contextual labelling.

Normative data layer: logs dimension score distributions by role family, seniority level, and technical domain. Accumulates normative benchmarks over time. Supports role-benchmark configuration for hirers who have designated top-performer reference profiles.

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7. Defining Target Customers

7.1 Primary Target Customers

Tier 1: High-Volume Technical Recruiters. Enterprise and Scale-Up

Any organisation running more than 30 to 40 technical hires per year has a structural headcount cost in screening and first-round assessment that Basanite directly displaces. This includes high-growth technology companies at Series B and beyond with dedicated talent teams, technology divisions of large financial institutions and professional services firms, and management consultancies running graduate and experienced-hire pipelines.

For this customer, the primary commercial argument is headcount displacement. One Basanite deployment can replace the screening and first-round assessment work of one to three full-time recruiters. At £60,000 to £90,000 per recruiter fully loaded, the payback period on a Basanite contract is measured in months, not years. Quality improvement is the reason the legal and ethical case for the product holds; cost displacement is the reason it gets purchased.

Tier 2: Staffing Agencies and Recruitment Process Outsourcing Firms

Staffing agencies and RPO firms face acute margin pressure as the market commoditises. Their differentiation must come from the quality and defensibility of the assessment they provide. A Basanite-powered agency can offer enterprise clients a materially stronger assessment product at lower per-candidate cost than a human-led screening process, while improving gross margin by reducing recruiter hours per placement.

Tier 3: SMEs Without Dedicated HR Functions

Small and medium-sized technology companies, typically between 20 and 150 employees, often have no dedicated HR or recruiting function. For this customer, Basanite provides a capability that does not exist. The commercial framing is closer to infrastructure than labour substitution: Basanite gives a 30-person company the evaluation sophistication of a company ten times its size, at a fraction of the cost of hiring a Head of Talent.

Tier 4: Universities, Graduate Employers, and Professional Training Programmes

Universities and large graduate employers are running assessment centres that cost £500 to £2,000 per candidate in assessor time, venue costs, and coordination overhead. Basanite can execute the first two to three stages of that funnel at a fraction of the cost, with demonstrably superior consistency.

Tier 5: Geographically Distributed and Remote-First Organisations

Asynchronous AI-led evaluation removes the coordination friction of human-led assessments across time zones. For a company hiring engineers across three continents from a central talent function, the operational value of an always-available, timezone-agnostic assessment system is significant and immediate.

7.2 Who Basanite Is Not For At This Stage

Non-technical roles, at least initially. Basanite's assessment architecture is purpose-built for technical capability evaluation. Extending to sales, operations, legal, or marketing roles would require a substantive rebuild of the core assessment logic.

Roles where the evaluation is primarily relational or political. Senior executive hiring is governed by dynamics that no structured assessment system currently handles well. These decisions involve stakeholder alignment, organisational politics, and relationship capital that are genuinely resistant to systematic evaluation.

Organisations that have decided they want a human face on every candidate interaction. Some companies have made a deliberate values choice that every hiring touchpoint will be human-led. This is a values position, not a capability gap, and is not one Basanite should attempt to argue against.

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8. User Journeys

8.1 The Hirer's Journey

Stage 1: Account Setup and Job Configuration

The hirer logs into their Basanite account and navigates to the job creation flow. They paste or compose a job description in the same format they would use on any standard hiring platform: role title, responsibilities, requirements, and contextual information about the team or company.

Basanite's configuration agent analyses the job description and performs an ONET role-family mapping. It surfaces a recommended evaluation dimension set drawn from Basanite's Layer 1 dimension library, with a plain-language rationale for each dimension. For a senior backend engineer role, the system might recommend judgment under ambiguity, sound intuition under data scarcity, and technical depth at the architecture register. For a junior data analyst, it might weight creative problem reframing and applied technical depth differently.

The hirer can accept the recommendation as a complete package, remove dimensions, or add dimensions from the full library. Each dimension is presented with its ONET anchor, a plain-language description, and example signal indicators.

Stage 2: Role Benchmark Configuration (optional)

For hirers with established teams, Basanite offers a role benchmark configuration. The hirer designates two to three current top performers in the target role. Basanite generates an internal benchmark profile from their assessment data or from a structured reference input. Subsequent candidates are scored relative to this role-specific benchmark in addition to the absolute BARS rubrics, producing a signal that reflects what good looks like in this specific organisational context rather than a generic population norm.

Stage 3: Eligibility Constraints and Budget

Basanite prompts the hirer to specify hard eligibility constraints (right to work, graduate status, minimum experience, technical prerequisites, location) stored as pre-screening filters. Basanite then asks the hirer to specify their assessment budget, which determines the depth and scope of assessment each candidate receives.

Stage 4: Role-Specific Prompt Generation

With job description, dimensions, constraints, and budget confirmed, Basanite generates a role-specific interview agent configuration. This includes the CTAI construct templates for each active dimension, the BARS rubrics, the follow-up trigger conditions, and the technical depth register. The hirer can preview the types of questions and assessment areas in plain language; the specific template logic remains internal to protect assessment integrity. The role is now live.

8.2 The Candidate's Journey

Stage 1: Application and Onboarding

The candidate arrives at the Basanite assessment link. They create or log into their Basanite account. Previous assessment history is retained in their profile, but each assessment is conducted fresh against the specific role's parameters.

The candidate uploads their CV. Basanite processes this document and extracts experience history, technical background, project work, and signals relevant to the role. This information populates the context slots in the CTAI templates for this candidate's specific interview. Before the assessment begins, Basanite presents the candidate with a plain-language explanation of what the assessment involves, approximately how long it will take, and what kinds of questions they can expect. Candidates are told they will be asked about their real experiences and their genuine ways of thinking, and that there are no trick questions or expected answers they should be trying to perform toward. They are also informed that the hirer may receive contextual observations in addition to their evaluation score.

Stage 2: Assessment

The candidate enters the interview. Basanite begins with an open invitation for the candidate to walk through a recent piece of work in their own words. This narrative foundation establishes a shared factual basis for follow-up questions, gives the candidate a comfortable entry point, and provides the system with an initial profile of experience depth and communication style against which to calibrate the remainder of the assessment.

From this foundation, Basanite moves through the active Layer 1 dimensions. Questions are generated from CTAI construct templates with context slots filled from the candidate's CV. Where the candidate has demonstrated relevant prior experience, the system deploys experience-anchored questions. Where relevant experience is absent or thin, the system activates open sandbox questions designed to surface reasoning and judgment in hypothetical but realistic scenarios.

Throughout the interview, signal confidence monitors track whether the BARS rubric can be scored with confidence for each dimension. When the candidate's response raises authenticity flags, the appropriate verification probe from the follow-up logic fires. The interview continues until signal confidence thresholds are met across the required dimensions or the budget-determined time limit is reached.

The candidate experience throughout is designed to feel like a substantive, curious conversation. Basanite does not rush, does not deploy trick questions, and does not penalise candidates for acknowledging uncertainty. Candidates who say "I don't know" in a way that demonstrates genuine metacognitive awareness are treated differently from candidates who produce confident but hollow answers.

Stage 3: Dual Feedback Reports

When the assessment concludes, Basanite generates two separate outputs.

The **hirer report** contains dimension-by-dimension Layer 1 scores with verbatim candidate statements cited as evidence for each score, a cheating risk assessment, a technical capability map identifying areas of depth and potential blind spots, a Layer 2 contextual observations section (clearly labelled as non-scored), and a one-paragraph synthesis of what this candidate is likely to be like in a real working environment. The report is designed to function as a briefing document for the final human-led interview: it tells the interviewer where to probe further, what has already been established, and what remains genuinely uncertain. It does not make a hire or no-hire recommendation.

The **candidate report** is a brief, neutral summary in plain language acknowledging what they demonstrated well, identifying areas for development, and offering constructive suggestions. The candidate report does not contain Layer 2 content and does not reveal the specific evaluation dimensions used, the scoring criteria, or the question logic. A candidate cannot use their feedback to game a subsequent Basanite assessment.

Stage 4: Ranking and Queue Management

The candidate is assigned a position in the hirer's live ranking queue. The composite score draws on Layer 1 dimension scores, background signals from the CV, stated salary expectations, and hard eligibility data. The ranking is dynamic: as new candidates complete assessments, the queue updates. The hirer can sort and filter by dimension, salary band, experience level, or composite score. When moving candidates to final human-led interview, the hirer does so with Basanite's full assessment in hand.

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9. Technical Innovation and Differentiation

9.1 Basanite Interview Technique Inventory

9.1.1 Structural Techniques

CTAI Template Architecture Every interview question is generated from a pre-specified construct template. The construct scenario, BARS rubric, and follow-up logic are invariant across candidates. The context slot is populated from the candidate's CV. This is the mechanism by which Basanite achieves construct-level standardisation with surface-level personalisation.

Macro Three-Part Structure Every interview follows a fixed high-level arc: narrative foundation → dimension assessment → verification probes (conditional). Candidates must establish their own story before probing begins.

Signal Confidence Pacing Transitions between dimensions are governed by internal confidence thresholds, not question counts. The interviewer moves on when sufficient BARS evidence has been accumulated, not when a quota is met.

Experience Path Bifurcation The interview forks based on whether the candidate has relevant prior experience. Path A deploys experience-anchored questions from CTAI templates. Path B deploys open sandbox questions with a weighted composite scoring model.

Technical Depth Calibration Before the interview begins, the agent calibrates to one of two depth registers (applied or research/architecture) based on the job description.

9.1.2 Core Questioning Techniques

Narrative Anchoring The opening phase invites the candidate to describe a recent piece of work in their own words, establishing a shared factual baseline from which all subsequent questions grow organically.

CTAI Template Questions Questions are generated from construct templates with CV context slots filled. The construct being elicited is identical across all candidates for a given dimension; the contextual framing is personalised.

Open Sandbox Questioning Where real experience is absent or insufficient, the candidate is placed in a realistic hypothetical scenario. Five specialist sandbox design principles govern these questions to prevent them from being answerable by AI lookup alone.

Decision Tracing The candidate is asked not only what they chose but why they did not choose the alternatives. The quality of their dismissal of Option B reveals more than their justification of Option A.

Boundary Condition Probing Rather than asking whether a solution works, the interviewer asks under what conditions it fails. Valid technical knowledge has edges; performed knowledge does not.

Compulsory Trade-Off Framing Sandbox questions are structured to make "consider all dimensions" a wrong answer strategy. Resource constraints or mutually exclusive choices are built into the scenario to force prioritisation.

9.1.3 Verification Probe Techniques

This part needs to be refined on further research!

These are deployed conditionally when trigger conditions are met (see Section 4.5). They are not part of the standard interview flow for candidates whose initial answers meet signal confidence thresholds.

Counterfactual Pressure In the verification phase, the candidate is asked what would have happened if they had made the opposite decision. This requires sustained reasoning in a direction for which they have no prepared answer.

Forced Commitment When a candidate gives a framework-level answer, the interviewer demands a specific, defensible position. This disrupts the tendency toward balanced non-answers.

Progressive Excavation Each layer of follow-up goes one level deeper into the same territory rather than expanding to new ground. The goal is vertical depth, not horizontal coverage.

Vagueness Targeting When a candidate gives an answer that sounds complete, the interviewer identifies the single most ambiguous detail within it and asks the candidate to elaborate specifically on that point.

Fourth Wall Break Mid-scenario, the interviewer pulls the candidate out of the hypothetical and asks what was genuinely the first thing that came to mind: not the answer they organised, but the raw initial reaction. AI has no first reaction; humans do.

Honest Failure Elicitation The candidate is asked to name a decision in their work that they did not think through fully but that happened to turn out fine anyway. Clean answers to this question are a high-risk signal.

Predict-Your-Own-Error After a candidate gives a judgment, they are asked where that judgment is most likely to be wrong based on their own known patterns of error.

Regret Probing For any experience the candidate describes as a success, the interviewer asks what they would have done differently. Authentic experience always contains regret; performed experience does not.

9.1.4 Consistency and Authenticity Verification Techniques

This part needs to be refined on further research!

Narrative Consistency Tracking All specific details mentioned by the candidate are internally logged and referenced in later questions to test whether the account holds together across the full conversation.

Parameter Verification For technical claims, the interviewer asks for concrete numbers: dataset sizes, latency figures, team headcount, timeline specifics.

Cognitive Priority Testing A key detail is buried in a scenario description, one that an experienced practitioner would immediately identify as the most important signal, but that an inexperienced candidate or AI would treat as background noise.

Information Gap Injection A critical variable in a scenario is deliberately left ambiguousone that only real experience can resolve through assumption.

9.1.5 Anti-Cheating Techniques

This part needs to be refined on further research!

Structural Opacity Evaluation dimensions are never disclosed to the candidate. Question purposes are concealed. The candidate cannot reverse-engineer what is being measured.

AI Output Signal Detection The interviewer maintains a checklist of linguistic and structural patterns associated with AI-generated responses: comprehensive coverage without prioritisation, clean narrative arcs without friction, self-criticism that reads as a growth story, absence of any moment of genuine uncertainty.

Cognitive Load Escalation The deepest verification probes are reserved for the areas where the candidate is most confident, not most vulnerable. Defences are lowest where people feel they are winning; that is where authentic signal is easiest to surface.

Latency Awareness Anomalous response timing is flagged as a risk indicator.

Tacit Knowledge Consistency Test A candidate who claims a skill they cannot teach, but then describes its internal mechanism with unusual precision, has produced a self-contradicting account.

9.1.6 Scoring and Output Techniques

BARS-Anchored Scoring Every dimension is scored against a pre-specified Behaviourally Anchored Rating Scale derived from ONET occupational descriptor language. No dimension may receive a score above 3 without a specific verbatim statement from the candidate cited as evidence.

Cheating Risk as Independent Dimension Cheating risk is scored and reported separately from capability.

Dual Report Generation Two separate outputs are produced from the same assessment: a hirer report and a candidate report, each serving different communicative purposes.

Technical Capability Mapping Rather than a single technical score, the output produces a structured map: areas of demonstrated depth, areas of surface fluency without underlying understanding, specific nodes requiring human technical expert verification.

9.2 Normative Data Architecture

Every Basanite assessment generates structured data events: dimension scores, signal patterns, cheating risk indicators, and role-family metadata. From the first deployment, the data architecture logs these outputs in a schema designed to accumulate normative benchmarks over time: dimension score distributions by role family, seniority level, and technical domain.

The long-term objective is a proprietary normative database that allows Basanite to contextualise individual scores against a statistically robust reference population, analogous to the normative infrastructure that constitutes SHL's most defensible competitive asset. This database will not be mature at launch; it must be designed before launch.

9.3 Role Benchmark Feature

For hirers with established teams, the role benchmark feature allows candidates to be scored relative to an internal reference profile derived from designated top performers. This produces a personalised, role-specific signal calibrated to what good actually looks like in this organisation, rather than a generic population norm.

10. Validation Plans

10.1 Layered Validity Architecture

Basanite's validation programme is structured across four layers, each addressing a distinct type of validity evidence and building cumulatively toward the full predictive validity case.

Layer 1: Construct Validity

Does Basanite measure what it claims to measure? Expert panel calibration: recruit 15–20 domain experts to complete Basanite assessments; separately rate the same individuals on target constructs using independently developed BARS. Convergent validity analysis. Discriminant validity testing via factor analysis of dimension scores across a large candidate sample.

Layer 2: Content Validity

Is the assessment sampling the right domain? Structured panel of I-O psychologists and senior technical practitioners produces content validity ratios (Lawshe 1975 methodology) for each dimension.

Layer 3: Concurrent Validity

Do Basanite scores correlate with expert assessment of the same individuals? Administered to known high and low performers in target roles, with assessor-blind scoring.

Layer 4: Predictive Validity

Do Basanite scores predict downstream job performance? Multi-source performance criterion at 6 and 12 months: structured peer assessments, objective output measures where available, manager ratings with forced distribution and behavioural anchoring, and performance trajectory over time.

All validity studies will be pre-registered on the Open Science Framework before data collection begins.

10.2 CTAI Validity Position

Basanite will commission or partner on research specifically examining how AI-assisted candidate preparation has affected the predictive validity of fixed-question structured interviews. This research addresses the mechanism by which Level 4 structured interviews have lost validity in the current environment and will provide the empirical basis for Basanite's methodological claims.

10.3 Adverse Impact Monitoring

From the first deployment, all assessment outputs are logged against voluntary demographic data (gender, ethnicity, age group). Quarterly analyses apply the EEOC 4/5ths rule to check pass rate ratios across demographic groups. A pre-specified remediation pathway exists for each dimension in the event that disparate impact is detected. This monitoring applies to both Layer 1 scored dimensions and Layer 2 observations. Results are available to enterprise clients on request as part of procurement documentation.

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11. Academic Partnerships

[To be completed.]

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12. Corporation Partnerships

[To be completed.]

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13. FAQ

This section addresses the most common questions raised by prospective customers, investors, and implementation partners. It is designed to be honest about both the product's genuine advantages and the areas where Basanite's approach involves deliberate trade-offs.

For Potential Customers (Put On Website!)

Q: How is this different from a structured interview?

Basanite uses a methodology we call Construct-Templated Adaptive Interviewing (CTAI), which achieves construct-level standardisation rather than item-level standardisation. In a conventional structured interview, every candidate receives the same question word for word. In CTAI, every candidate receives a question generated from the same construct template. The cognitive demand, scoring rubric, and follow-up logic are identical, but the surface content is personalised to the candidate's own CV and experience.

The psychometric analogy is parallel test forms: Form A and Form B of a cognitive ability test contain different items but are considered equivalent instruments because they elicit identical constructs under identical scoring. CTAI generates the equivalent of a bespoke parallel form for each candidate.

This approach is not a compromise relative to conventional structured interviewing. It preserves the validity-critical property (consistent construct elicitation and evaluation) while eliminating the validity-degrading property of fixed question sets: their susceptibility to coaching and AI-assisted preparation.

Q: Can candidates game the system by over-preparing?

Significantly less so than with any fixed-question system. Because each candidate receives questions drawn from their own CV rather than a shared question bank, there is no shared question set to prepare for. A candidate who searches "Basanite interview questions" will not find the questions they will be asked, because those questions do not exist until their CV is processed.

Preparation that focuses on understanding a topic deeply and being able to discuss real experiences is not gaming. Preparation that involves memorising scripted answers to known questions is precisely what CTAI prevents.

Q: Is the system biased? How do you ensure fairness?

Basanite runs quarterly adverse impact analyses using the EEOC 4/5ths rule, monitoring pass rate ratios across demographic groups for all scored dimensions. Results are available to enterprise clients on request. The CTAI approach reduces one well-documented source of bias in conventional structured interviews: the advantage that accrues to candidates who are culturally familiar with the expected answers to generic questions. By anchoring questions in each candidate's own experience, the system creates conditions where a self-taught engineer without institutional pedigree encounters the same construct, the same rubric, and the same depth of follow-up as one from a target university.

Q: What does the hirer actually receive at the end?

A structured assessment report containing: dimension-by-dimension scores with verbatim candidate statements cited as evidence; a cheating risk assessment scored independently from capability; a technical capability map distinguishing demonstrated depth from surface fluency; a contextual observations section (clearly labelled as non-scored); and a synthesis paragraph describing what this candidate is likely to be like in a real working environment. The report is designed to function as a briefing document for the human final interview. It tells the interviewer where to probe further, what has already been established, and what remains genuinely uncertain. It does not make a hire or no-hire recommendation.

Q: What does the candidate receive?

A brief, neutral summary acknowledging what they demonstrated well, identifying areas for development, and offering constructive suggestions. Candidate reports are designed to be informative without being reverse-engineerable, they do not reveal the specific dimensions, scoring criteria, or question logic. A candidate cannot use their feedback to game a subsequent Basanite assessment. Every candidate should leave the process with something of genuine value regardless of outcome.

Q: How long does the interview take?

Interview length is determined by the budget tier confirmed at role setup and by signal confidence accumulation. Most assessments fall in the range of 25 to 60 minutes. Candidates whose answers accumulate signal confidence quickly will have shorter interviews; candidates whose answers trigger verification probes will encounter additional depth. The interview does not continue beyond the budget-determined limit.

Q: Does Basanite make the hiring decision?

No. Basanite produces assessment outputs that support the hirer's decision. The hire or no-hire recommendation remains with the human. The hirer report is explicitly designed as a briefing document for the human final interview, not as an autonomous decision mechanism.

Q: Does it work for all technical roles?

The current architecture is optimised for software engineering, data, and adjacent technical roles where the judgment and reasoning constructs in the dimension library are most directly relevant. The configuration agent's ONET role-mapping supports a range of role families within technical hiring. Non-technical roles are not currently supported and would require a substantive rebuild of the core assessment logic.

For Investors

Q: What is the core competitive moat?

Three compounding advantages that are difficult to replicate independently and that reinforce each other over time.

First, the methodology itself. CTAI with AI-consistency produces assessments that are simultaneously more resistant to gaming, more ecologically valid, and more consistently delivered than any legacy tool. This is a structural capability advantage that cannot be replicated by adding AI features to a static question bank.

Second, normative data. Every assessment Basanite conducts generates dimension score data that feeds the normative benchmarking system. As the dataset grows, Basanite's ability to contextualise individual scores against statistically robust role-family benchmarks becomes more precise and more defensible. SHL's 35 million annual assessments constitute its most defensible competitive asset; the architecture to build an equivalent moat must be designed from Day 1.

Third, the authenticity signal. No legacy tool has a mechanism to distinguish genuine capability from AI-assisted performance. Basanite's construct verification approach is the only architecture in the market that produces this signal systematically. As AI assistance in candidate preparation becomes ubiquitous, this capability becomes increasingly load-bearing for any hiring system.

Q: How large is the addressable market?

The primary commercial case rests on headcount displacement. Any organisation running more than 30–40 technical hires per year has a structural recruiter cost in screening and first-round assessment that Basanite directly displaces. At £60,000–90,000 per recruiter fully loaded, one Basanite deployment can reduce costs by £60,000–270,000 per year for the target customer. The market includes high-growth technology companies, enterprise technology divisions, staffing agencies and RPO firms, and universities and graduate employers, all of whom are running assessment infrastructure that is both expensive and increasingly invalid.

Q: Where does validity evidence currently stand?

The current CTAI methodology has the theoretical grounding of parallel test forms psychometrics and Campion et al.'s research on evaluation structure as the primary validity contributor in structured interviews. Full predictive validity studies, the gold standard in I-O psychology, require deployment data and are a multi-year programme. The validation architecture described in Section 10 is designed to build this evidence in stages, with all studies pre-registered before data collection begins. Enterprise procurement at regulated institutions will require this evidence; the roadmap is designed to produce it in the sequence that matters commercially.

Q: What is the regulatory and legal risk?

Assessment tools used in hiring decisions are subject to employment discrimination law in most jurisdictions. Basanite's primary risk mitigations are: ONET-anchored dimensions provide construct validity by inheritance from validated occupational science; BARS-anchored scoring with verbatim citation creates an auditable evidence record for every score; adverse impact monitoring from Day 1 identifies demographic disparities before they become systemic; and Layer 2 observations are contractually restricted from use as hiring criteria. The risk profile is substantially lower than an unvalidated bespoke assessment and substantially lower than an unstructured interview.

For Academic and Research Partners

Q: How does CTAI relate to the structured interview literature?

CTAI is a methodologically distinct approach that should not be characterised as a structured interview in the Campion et al. sense, and we do not claim that it is. The conventional structured interview literature defines structure as requiring identical questions. CTAI achieves construct-level standardisation through pre-specified templates and BARS rubrics without item-level standardisation.

The theoretical foundation for treating this as a valid approach is twofold. Campion et al. (1997) established that evaluation structure: standardisation of scoring rather than question content, is the more validity-predictive of the two dimensions of interview structure. Additionally, the parallel test forms principle in psychometrics establishes that instruments can be considered equivalent when they elicit identical constructs under identical scoring, even when surface items differ. CTAI is essentially an application of parallel test forms logic to the interview context, enabled by generative AI's capacity to produce construct-equivalent forms at scale.

We expect this to be contested by some psychometricians, and we welcome that challenge. The appropriate response is empirical: validity studies comparing CTAI outcomes to conventional structured interview outcomes, under conditions that account for the degradation of fixed-question validity through AI-assisted preparation. We are designing this research and welcome academic collaboration on its execution.

Q: What is the evidential status of the evaluation dimensions?

Layer 1 dimensions are mapped to ONET work activity codes and are scored using BARS rubrics derived from ONET occupational descriptor language. They have construct validity by inheritance from validated occupational science infrastructure.

Layer 2 observations are generated in parallel and do not contribute to candidate scoring. They represent constructs with theoretical grounding (Polanyi, Collins, Klein, Edmondson) but without psychometric validation history sufficient for scored evaluation.

We are explicit about this distinction in the product and in all external communications. The construct validity programme (Section 10.1) is designed to produce publishable evidence on whether Layer 1 dimensions are real, distinct, and coherent as scored constructs, and to provide the empirical basis for eventually graduating Layer 2 observations to scored status.

Q: Are you claiming to measure tacit knowledge?

We are making a weaker and more defensible claim: we are attempting to detect the surface signatures of tacit knowledge, the ability to name specific calibration factors, to identify what an experienced practitioner would notice as the critical variable, to describe the conditions under which a known approach fails. Whether these signatures constitute direct measurement of tacit knowledge in the Polanyian sense, or whether they constitute proxies that correlate with the underlying construct, is an empirical question we do not currently have the evidence to answer definitively. The construct validation programme is designed to address this. We have deliberately placed tacit knowledge surfacing in Layer 1 at a reduced confidence level rather than treating it as fully validated, because intellectual honesty about capability boundaries is a design requirement, not a marketing disclaimer.